

WEEK 3 -THEORY

UI Architecture - Theme, Widgets, Screens

Learning objectives

- ✓ **Structure Flutter widgets** for extendibility and consistency
- ✓ Comply with **coding conventions**
- ✓ Create a library of **generic widgets** aligned with a **design system**
- ✓ Understand the dart concepts of **static methods, attributes**

How to start?

- ✓ Create a **GitHub repository** for this project
- ✓ Get the **start code**, including the Figma Design System
- ✓ Ensure you can run the start project on your computer

How to submit?

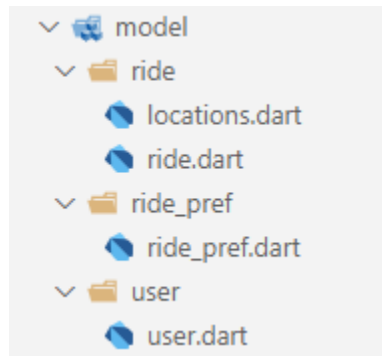
- ✓ Make sure you follow the COMMIT standards (see below)
- ✓ **Push the start code** into your repository (commit: BLA-000- Start Code)
- ✓ Once finished, submit to MS Team:
 - GitHub URL
 - This document

You are **not** allowed to **use AI** to submit this work

ACTIVITY 1 – Analyze the **model layer**

The model layer - so far - is composed of 6 classes:

Ride, Location, Country, DateTime, User, RidePref

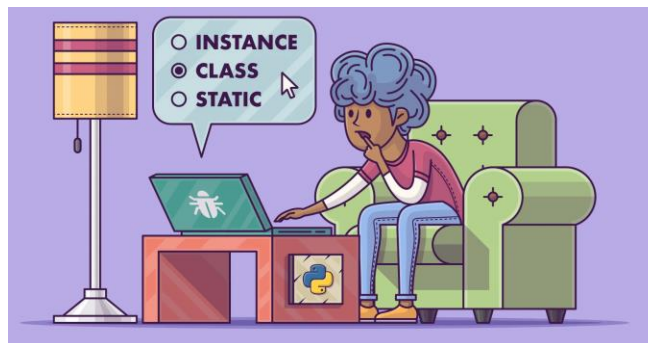


 Draw the **UML class diagram** representing the relationships between those 6 classes.

(no attribute, methods, just associations relationships)

 What is the goal of the operator `==` and the `hashCode` method in Location class?

ACTIVITY 2 – Understand the difference between **instance** and **static** access

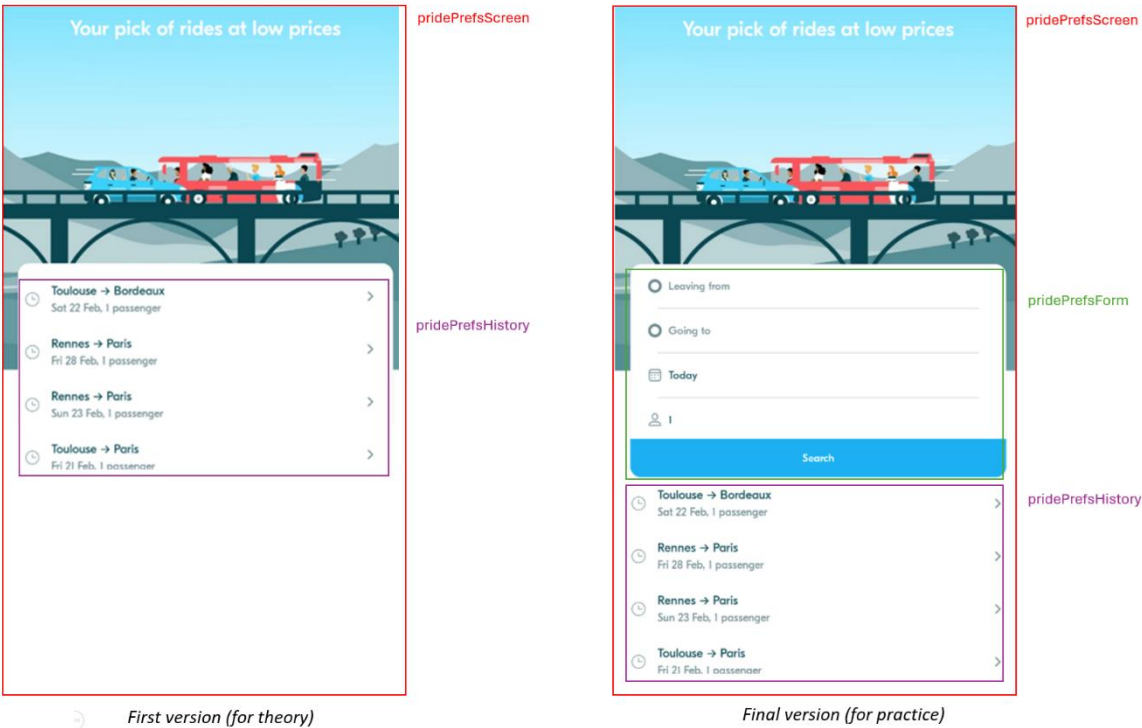


Answer the quiz in class

ACTIVITY 3 – Understand and analyze a first view: Ride Preferences screen

Let’s build step by step a Ride Preferences screen, showing only the past entered ride preferences.

Understand the methodology, steps, and coding standards to implement this view



Explain how the font (*Eesti*) is loaded from the assets and is applied to all widgets.

Analyze how the history tile interacts with the App theme.

History Tile	BlaTextStyle	BlaColor
title		
subtile		

Explain how the date is converted into a readable label

 Analyze the Ride Preferences screen and complete the table.

Widget	Screen / Screen Widget /App Widget	Parameters	Callbacks
RidePrefScreen			
BlaBackground			
RidePrefHistoryTile			

 At home, try to **reproduce it by yourself** (with the correction for help)